

```
#include <stdio.h>

#define MAX 5

int queue[MAX];

int front = -1, rear = -1, count = 0;

void insertRear(int val) {
    if (count == MAX) {
        printf("Queue is full\n");
        return;}

    if (front == -1) front = 0;

    rear = (rear + 1) % MAX;

    queue[rear] = val;

    count++;
}

void insertFront(int val) {
    if (count == MAX) {
        printf("Queue is full\n");
        return;}

    if (front == -1) {
        front = rear = 0;}

    else {
        front = (front - 1 + MAX) % MAX;}

    queue[front] = val;

    count++;
}

void deleteFront() {
    if (count == 0) {
```

```

        printf("Queue is empty\n");
        return;
    }

    printf("Deleted from front: %d\n", queue[front]);
    front = (front + 1) % MAX;
    count--;
    if (count == 0) front = rear = -1;
}

void deleteRear() {
    if (count == 0) {
        printf("Queue is empty\n");
        return;
    }

    printf("Deleted from rear: %d\n", queue[rear]);
    rear = (rear - 1 + MAX) % MAX;
    count--;
    if (count == 0) front = rear = -1;
}

void display() {
    if (count == 0) {
        printf("Queue is empty\n");
        return;
    }

    printf("Queue elements: ");
    for (int i = 0; i < count; i++) {
        int index = (front + i) % MAX;

```

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        printf("%d ", queue[index]);
    }
    printf("\n");
}

int main() {
    int choice, val;
    while (1) {
        printf("\n--- queue operation ---\n");
        printf("1. Insert at Rear\n");
        printf("2. Insert at Front\n");
        printf("3. Delete from Front\n");
        printf("4. Delete from Rear\n");
        printf("5. Display\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value: ");
                scanf("%d", &val);
                insertRear(val);
                break;
            case 2:
                printf("Enter value: ");
                scanf("%d", &val);
                insertFront(val);

```

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        break;
    case 3:
        deleteFront();
        break;
    case 4:
        deleteRear();
        break;
    case 5:
        display();
        break;
    case 6:
        return 0;
    default:
        printf("Invalid choice\n");
    }
}
}
```

Output

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 1
Enter value: 10

--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 2
Enter value: 20

--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 1
Enter value: 5
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Queue elements: 20 10 5
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 2
Enter value: 30
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 1
Enter value: 45
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Queue elements: 30 20 10 5 45
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 2
Enter value: 3
Queue is full
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Queue elements: 30 20 10 5 45
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 3
Deleted from front: 30
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 4
Deleted from rear: 45
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Queue elements: 20 10 5
```

```
--- queue operation ---
1. Insert at Rear
2. Insert at Front
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 6
```

```
Process returned 0 (0x0)   execution time : 89.054 s
Press any key to continue.
```

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